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Prognostic Significance of Transient Ischemic Episodes: Response to Treatment Shows Improved Prognosis

Results of the Total Ischemic Burden Bisoprolol Study (TIBBS) Follow-Up

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Objectives. **The Total Ischemic Burden Bisoprolol Study (TIBBS) follow-up examined cardiac event rates in relation to transient ischemia and its treatment.**

Background. **It is unclear whether transient ischemia on the ambulatory electrocardiogram has prognostic implications in stable angina and whether medical treatment can improve the prognosis.**

Methods. **The TIBBS trial was an 8-week, randomized, controlled comparison of the effects of bisoprolol and nifedipine on transient ischemic episodes in patients with stable angina pectoris. Of the 545 patients screened, 520 (95.4%) could be followed up. Rates of cardiac and noncardiac death, nonfatal acute myocardial infarction, hospital admission for unstable angina and need for coronary artery bypass graft surgery or percutaneous transluminal coronary angioplasty were recorded.**

Results. **A total of 145 events occurred in 120 (23.1%) of 520 patients. Patients with more than six episodes had an event rate of 32.5% compared with 25.0% for patients with two to six episodes and 13.2% for patients with less than two episodes ($p < 0.001$). Hard events (death, acute myocardial infarction, hospital admission for unstable angina pectoris) were more frequent in patients with two or more ischemic episodes (12.1% vs. 4.7%, $p = 0.0049$). Patients with a 100% response rate of transient ischemic episodes during the TIBBS trial had a 17.5% event rate at 1 year compared with 32.3% for non-100% responders ($p = 0.008$). Patients receiving bisoprolol during the TIBBS trial had a lower event rate (22.1%) at 1 year than patients randomized to nifedipine (33.1%, $p = 0.033$).**

Conclusions. **In patients with stable angina pectoris, frequent episodes of transient ischemia are a marker for an increased event rate. A 100% response to medical treatment reduces the event rate. The greater reduction of ischemia with bisoprolol than nifedipine during the TIBBS trial translated into an improved outcome at 1 year.**

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In different subsets of patients with coronary artery disease, prognostic significance has been demonstrated for transient ischemic episodes detected by ambulatory monitoring.

The highest correlation of transient ischemia with subsequent higher event rates was found for patients with unstable angina pectoris (1–9); however, patients who have had an acute myocardial infarction (10–14) and patients with stable angina pectoris (15–19) have also been shown to have increased event rates when transient, predominantly silent ischemia is present on ambulatory electrocardiographic (ECG) monitoring. In stable angina pectoris, the practical value of the ambulatory ECG for the detection of ischemia has been a matter of dispute (20–23), but recently published trials show an improved outcome after response to treatment (24,25).

The Total Ischemic Burden Bisoprolol Study (TIBBS) (26) was designed to compare the beta$_1$-selective blocking agent bisoprolol with the calcium channel blocking agent nifedipine, which were used to treat transient ischemia in a large patient group. Both drugs reduced the number of ischemic episodes, but bisoprolol proved to be significantly more effective. The follow-up study aimed to answer two questions: 1) Is a prognostic influence of transient ischemia demonstrable at 1 year; and 2) does the effect of medical treatment in reducing ischemia translate into an improved prognosis?

Methods

Initial study design. The follow-up to TIBBS was an observational study aimed at prognostic comparison in well-defined groups of TIBBS patients with stable angina pectoris and a positive exercise ECG. The design and results of TIBBS have been published in detail in this journal (26). In short, it was a randomized, double-blind, controlled, parallel group trial of bisoprolol (4 weeks at 10 mg once daily, followed by 4 weeks at 20 mg once daily) versus slow-release nifedipine (20 mg twice daily). It included patients with stable angina pectoris, positive exercise ECG results and two or more transient ischemic episodes on the 48-h ambulatory ECG. Both treat-

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Table 1. TIBBS Follow-Up: Number of Events (one or more event/patient)

	Cardiac Death	Noncardiac Death	MI	Hospital Admission for Unstable Angina	CABG	PTCA
All patients (n = 520)	4	2	12	35	72	20
Only 1 (severest) event/patient	4	2	11	32	55	16
Ischemic episodes at baseline						
<2 (n = 190)	0	0	2	6	13	5
2–6 (n = 164)	2	1	3	19	23	5
>6 (n = 166)	0	0	7	10	36	10
Responders						
100% (n = 97)	1	0	1	6	10	3
Non-100% (n = 186)	1	1	3	16	42	8
Bisoprolol (n = 154)	1	0	2	10	22	5
Nifedipine (slow release) (n = 163)	0	2	7	16	36	7

CABG = coronary artery bypass graft surgery; MI = myocardial infarction; PTCA = percutaneous transluminal coronary angioplasty; TIBBS = Total Ischemic Burden Bisoprolol Study.

ments greatly reduced the number of transient ischemic episodes, but bisoprolol was significantly more effective (e.g., 10 mg of bisoprolol reduced the number of transient ischemic episodes from a mean of 8.1 [SEM] $\pm$ 0.56 at 48 h to 3.2 $\pm$ 0.41, vs. nifedipine, 2 $\times$ 20 mg, from 8.3 $\pm$ 0.5 to 5.9 $\pm$ 0.43, $p < 0.001$). The randomized treatment phase of the study was completed in 8 weeks. After that, the patients' symptoms were treated at the discretion of their physicians, to whom the results of the ambulatory monitoring during treatment were not disclosed.

Follow-up. At 6 months and 1 year from the randomization, screened patients underwent follow-up examinations. Patients who were screened on the basis of a history of stable angina and a positive exercise ECG but who did not show two or more transient ischemic episodes, as well as those who showed transient ischemia and had been randomized to controlled treatment in TIBBS, were followed up.

Patients were excluded from follow-up if their 48-h Holter tapes were not technically acceptable during the placebo prephase of TIBBS. In all 30 participating centers, the patients were monitored by repeated visits, telephone calls or interviews with their treating physicians. Information was obtained as to whether the patients had had any of the following events: 1) cardiac death, 2) noncardiac death, 3) nonfatal myocardial infarction, 4) admission to the hospital owing to unstable angina, 5) the need for percutaneous transluminal coronary angioplasty or 6) coronary artery bypass graft surgery. Events 1 through 4 were evaluated together with the other events and separately as "hard events" because they do not have the bias problem of an elective treatment decision. The definition of nonfatal acute myocardial infarction followed standard criteria. Indications for admission owing to unstable angina and for revascularization were at the discretion of the physicians in charge of the patients in the different centers. These physicians were unaware of the results of the initial ambulatory ECG monitoring and the treatment assignment. Furthermore, patients were asked about the antianginal medication they had taken from the randomized treatment phase up to 1 year.

On the basis of history and positive exercise ECG results, 630 patients were included in the screening phase of TIBBS and underwent 48-h Holter monitoring. Of these patients, 85 had no or technically unacceptable Holter tapes during the placebo prephase and were excluded. Follow-up information was not available for 25 patients, thus giving a follow-up rate of 95.4%. Of the 520 patients evaluated for follow-up information, 203 were screened only by Holter ECG but not randomized, and 317 entered the randomized treatment phase of TIBBS. Of the latter group, a subgroup of 283 patients showed valid 48-h Holter tapes also during the TIBBS randomized treatment phase, and thus could be evaluated for treatment responses of transient ischemia.

Statistical analysis. All data from the follow-up report forms were continuously entered into a central data base. This allowed cross-correlations with information obtained on the same patients during TIBBS.

For comparisons of proportions of patients with events, the Fisher exact test or chi-square test was used, and the log-rank test was used for comparisons of event-free survival curves. The occurrence of events in the time course was described by life-table curves (Kaplan-Meier). Significance level was alpha 5% (two-tailed).

Results

Event rates. Table 1 shows the number of patients with events that occurred in our patient group. Of the 520 patients at risk, 120 had a total of 145 events. If only the severest event per patient was counted, the event rate was 23.1% (120 of 520), with 9.4% (49 of 520) hard events (death, acute myocardial infarction, hospital admission owing to unstable angina).

Medical treatment at 6 months of follow-up included platelet aggregation inhibitors (59.4%), beta-blockers (59.4%), long-acting nitrates (57.2%), short-acting nitrates (48.9%), calcium antagonists (47.8%), angiotensin-converting enzyme inhibitors (7.6%) and anticoagulants (7.0%). Beta-blockers were used exclusively (without calcium antagonists) in 37.3%

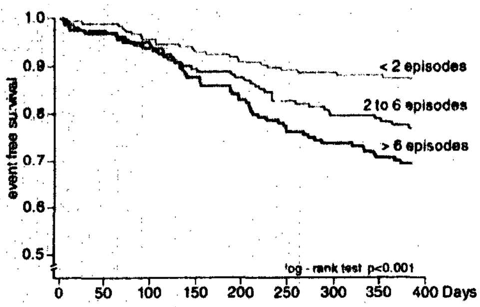


Figure 1. Event-free survival to first event in 520 patients screened for the Total Ischemic Burden Bisoprolol Study. One hundred ninety patients showed less than two transient ischemic episodes on 48-h ambulatory electrocardiographic monitoring; 164 patients had two to six episodes; and 166 patients had more than six episodes. A higher number of episodes is clearly followed by an increased event rate.

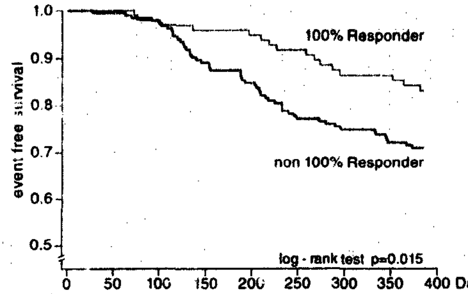


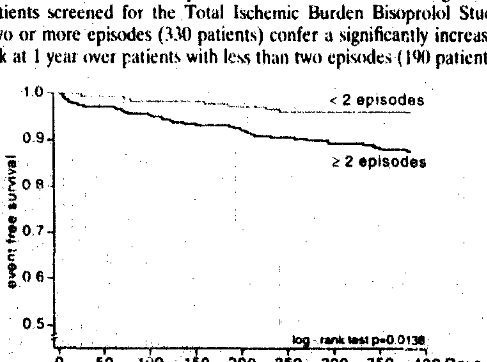
Figure 3. Event-free survival in 283 patients treated in the randomized, controlled phase of the Total Ischemic Burden Bisoprolol Study (only patients who had technically acceptable tapes throughout the 8-week trial). A 100% reduction of transient ischemic episodes was achieved in 97 patients who had a reduced risk of subsequent events over non-100% responders (186 patients).

of patients, calcium antagonists exclusively (without beta-blockers) in 27.9% and a combination of both in 19.8%. At 1 year of follow-up, there were no significant changes in the frequencies of the different medications used.

Prognostic influence of transient ischemia. Of the 190 patients with less than two ischemic episodes, 13.2% had an event, compared with 25% of the 164 patients with two to six episodes and 32.5% of the 166 patients with more than six episodes (chi-square, $p < 0.001$). Hard events occurred in 4.7%, 14.0% and 10.2% of those with less than two ischemic episodes, two to six episodes and more than six episodes, respectively ($p = 0.011$). Figure 1 shows the event-free survival for patients grouped according to the number of transient ischemic episodes on the 48-h Holter ECG during the placebo prephase of the TIBBS trial. There is a clear stepwise separation of the event-free probability curves between patients with less than two episodes and the other two groups with more severe ischemia.

In Figure 2 only hard events were counted for patients separated into two groups, those with less than two transient

Figure 2. Event-free survival for hard events only (death, acute myocardial infarction, hospital admission for unstable angina) in patients screened for the Total Ischemic Burden Bisoprolol Study. Two or more episodes (330 patients) confer a significantly increased risk at 1 year over patients with less than two episodes (190 patients).



ischemic episodes and those with two or more episodes. Again, there is a significant separation at 1 year, demonstrating the prognostic influence of transient ischemia. When the total ischemic burden and the duration of ischemic episodes are taken as measures of ischemia and the two groups are separated, respectively, on both sides of the median, then the event rates are also significantly higher in the groups with a higher total ischemic burden and a longer duration of ischemia.

Prognostic influence of response to treatment. Patients who showed a 100% reduction of their transient ischemic episodes under treatment had an event rate of 17.5%, compared with a 32.3% event rate in non-100% responders ($p = 0.008$). Hard events were 7.2% and 11.3% in 100% and non-100% responders, respectively ($p = 0.276$). Figure 3 shows the event-free survival curves for patients with or without a 100% response to medical treatment, i.e., no transient ischemic episodes on the control Holter tapes (week 4 or 8 of randomized treatment) during TIBBS. Patients who had a complete response and who lost all transient ischemia had a significantly improved outcome compared with the rest of the randomized patients. When only hard events were counted, the separation was not statistically significant (log-rank test, $p = 0.258$). Treatment responses with a <100% reduction of ischemic episodes did not show an improved prognosis. In fact, the event rates for patients with at least a 25% reduction (26.1%), at least a 50% reduction (24.7%) and at least a 75% reduction (24.3%) were similar and different from patients with a 100% reduction (17.5%).

Prognostic influence of initial randomized treatment. Patients randomized to receive bisoprolol had an event rate of 22.1%, compared with 33.1% in the patients receiving nifedipine (Fisher exact test, $p = 0.033$). Figure 4 compares the event-free survival curves for patients randomized to receive either bisoprolol or nifedipine during the initial randomized treatment phase (8 weeks). Patients who had been given bisoprolol during TIBBS fared significantly better than those who had been taking nifedipine. Even when only the hard

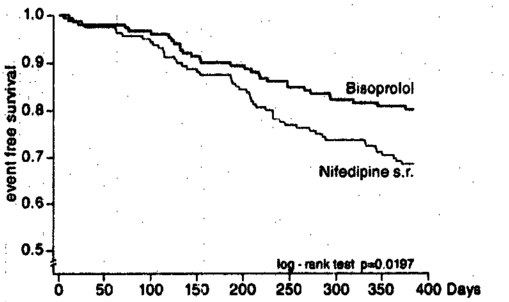


Figure 4. Event-free survival for 317 patients treated in the randomized, controlled phase of the Total Ischemic Burden Bisoprolol Study. The patients who had received bisoprolol (n = 154) during the trial had a significantly lower event rate than patients who had received slow release (s.r.) nifedipine (n = 163).

events were counted, a trend remained (log-rank test, p = 0.104).

Patients randomized to receive bisoprolol during the initial phase of the study were kept on beta-blockers by their physicians more often than patients randomized to receive nifedipine (47% vs. 32%, p = 0.008). Treatment with calcium antagonists during follow-up was comparable for both groups (21% vs. 26%, p = 0.29).

Discussion

Influence of patient selection. The present study followed patients who had undergone a screening phase for a randomized, controlled treatment study. Thus it could compare patients who had shown episodes of transient ischemia to others who had not. It has to be noted that all patients had a history of stable angina pectoris and positive exercise ECG results. Thus, the results pertain especially to the additive prognostic influence of transient ischemic episodes on ambulatory 48-h monitoring. The follow-up rate at 1 year was >95%, and thus the results validate earlier studies showing prognostic significance for transient, predominantly silent ischemia in patients with stable angina pectoris (15-19). Our study group was large enough to show a significant prognostic separation also for hard events.

Our results underscore the prognostic importance of transient ischemia, if demonstrated additively to a positive exercise ECG. Transient ischemic episodes may represent a sign of underlying instability of the disease in patients who otherwise present with stable angina pectoris. In the study of Mulcahy et al. (20), a positive exercise ECG was not a prerequisite for inclusion of the patients, and this "dilution" of the patient group may be the reason for the lack of prognostic implications in their study. We would, however, think that the prognostic influence of transient ischemia that we have shown is not only statistically significant but also clinically relevant.

Prognostic influence of treatment response. The follow-up of patients who had undergone the randomized treatment phase of TIBBS showed a remarkable effect of treatment response on prognosis. Although the treatment was not randomized over 1 year, but only for the 8 weeks of the trial, the results are valid, as the results of the ambulatory monitoring during the trial were not disclosed to the physicians in charge of the patients during following year. The present study is, of course, observational in this respect, but treatment responses on ambulatory ECGs were measured during the controlled, randomized phase of the study. The effect of initial treatment could only have been diluted during follow-up, especially for hard events. Our study shows that a complete treatment response confers an improved prognosis for the patients, even over 1 year. It is important that the response of ischemia be 100%, because lesser reductions of episodes were not associated with improved prognosis. Our results compare very well with the recently published Atenolol Silent Ischemia Study (ASIST) trial (24). In this study patients underwent randomized treatment with atenolol or placebo, and a beta-blocker also reduced the amount of ischemia and improved prognosis. As in our study, complete relief of ischemia under medical treatment was important for a positive influence on prognosis. In regard to the marked effect, which the treatment response during the 8-week trial had for the outcome at 1 year, the recent report of Gill et al. (27) is interesting in their comparison of 406 patients who had an acute myocardial infarction: those with ischemia on the ambulatory ECG had a highly increased event rate, the difference occurring almost completely during the first 3 months. Pepine (28) pointed out the possibility of biologically unstable coronary artery disease in a clinically stable patient.

Differences between treatments. In our study the positive treatment effects shown by bisoprolol during the randomized phase of the trial (26) translated into improved outcomes at 1 year. This seems difficult to explain at first, but the initial stabilization of some degree of instability may be one reason. Another important factor may be that the patients who had been randomized to bisoprolol treatment were significantly more likely to be kept on beta-blockers for the following year. Thus, the higher rate of complete treatment responses obtained with bisoprolol may have had an extended effect on subsequent prognosis. With the higher dose of 20 mg of bisoprolol a 100% response of ischemic episodes was achieved in 52% of the patients (26). Thus, a sizable proportion of patients who are probably not treated to an optimal outcome remains. Further studies will have to address the question of whether the percentage of complete treatment responses can be improved by combination therapy of a beta-blocker like bisoprolol with a calcium antagonist (29,30), aggressive lipid-lowering therapy (31) or a strategy including revascularization (25,32,33). The recently published Asymptomatic Cardiac Ischemia Pilot (ACIP) trial (25,32,33) showed revascularization to be somewhat superior to medical treatment in the reduction of subsequent cardiac events.

Conclusions. The 1-year follow-up of patients with stable angina pectoris, but demonstrable frequent transient ischemia in addition to positive exercise ECG results, showed a significantly worse prognosis than patients with less or no transient ischemia. This finding may represent a degree of hidden instability of the coronary artery disease in such patients. Medical treatment can improve the prognosis if it achieves a 100% reduction of ischemic episodes. Patients taking bisoprolol during the TIBBS trial fared better than those initially randomized to receive nifedipine, probably because patients taking bisoprolol had a higher rate of 100% responses.

References

1. Johnson SM, Mauritson DR, Winniford MD, et al. Continuous electrocardiographic monitoring in patients with unstable angina pectoris: identification of high-risk subgroup with severe coronary disease, variant angina, and/or impaired early prognosis. Am Heart J 1982;103:4–12.
2. Gottlieb SO, Weisfeldt ML, Ouyang P, et al. Silent ischemia as a marker for early unfavorable outcomes in patients with unstable angina. N Engl J Med 1986;314:1214–9.
3. Gottlieb SO, Weisfeldt ML, Ouyang P, Mellits ED, Gerstenblith G. Silent ischemia predicts infarction and death during 2-year follow-up of unstable angina. J Am Coll Cardiol 1987;10:756–60.
4. Nademanee K, Intarachot V, Josephson MA, Rieders D, Mody FV, Singh BN. Prognostic significance of silent myocardial ischemia in patients with unstable angina. J Am Coll Cardiol 1987;10:1–9.
5. von Arnim T, Gerbig HW, Krawietz W, Höfling B. Prognostic implications of transient predominantly silent ischaemia in patients with unstable angina pectoris. Eur Heart J 1988;9:435–40.
6. Langer A, Freeman MR, Armstrong PW. ST segment shift in unstable angina: pathophysiology and association with coronary anatomy and hospital outcome. J Am Coll Cardiol 1989;13:1495–1502.
7. Bugiardini R, Pozzati A, Borghi A, et al. Angiographic morphology in unstable angina and its relation to transient ischemia and hospital outcome. Am J Cardiol 1991;67:460–4.
8. Pozzati A, Bugiardini R, Borghi A, et al. Transient ischaemia refractory to conventional medical treatment in unstable angina: angiographic correlates and prognostic implications. Eur Heart J 1992;13:360–5.
9. Holdright D, Patel D, Cunningham D, et al. Comparison of the effect of heparin and aspirin versus aspirin alone on transient myocardial ischemia and in-hospital prognosis in patients with unstable angina. J Am Coll Cardiol 1994;24:39–45.
10. Tzivoni D, Gravish A, Zin D, et al. Prognostic significance of ischemic episodes with previous myocardial infarction. Am J Cardiol 1988;62:661–4.
11. Gottlieb SO, Gottlieb SH, Achuff SC, et al. Silent ischemia on Holter monitoring predicts mortality in high risk postinfarction patients. JAMA 1988;259:1030–5.
12. Langer A, Minkowitz J, Dorian P, et al., the TPAT study group. Pathophysiology and prognostic significance of Holter-detected ST segment depression after myocardial infarction. J Am Coll Cardiol 1992;20:1313–7.
13. Currie P, Ashby D, Saltissi S. Prognostic significance of transient myocardial ischemia on ambulatory monitoring after acute myocardial infarction. Am J Cardiol 1993;71:773–7.
14. Bonaduce D, Petretta M, Lanzillo T, et al. Prevalence and prognostic significance of silent myocardial ischemia detected by exercise test and continuous ECG monitoring after acute myocardial infarction. Eur Heart J 1991;12:186–93.
15. Deedwania P, Carbajal EV. Silent ischemia during daily life is an independent predictor of mortality in stable angina. Circulation 1990;81:748–56.
16. Yeung AC, Barry J, Orav J, Bonassin E, Rabyu KE, Selwyn AP. Effects of asymptomatic ischemia on long-term prognosis in chronic stable coronary disease. Circulation 1991;83:1598–604
17. von Arnim T, Szeimies-Seebach U, Erath A, Schreiber MA, Höfling B. Ischemia detected on ambulatory monitoring: has it prognostic implications? Adv Cardiol 1990;37:236–43.
18. Rocco MB, Nabel EG, Campbell S, et al. Prognostic importance of myocardial ischemia detected by ambulatory monitoring in patients with stable coronary artery disease. Circulation 1988;78:877–84.
19. Tzivoni D, Weisz G, Gavish A, Zin D, Keren A, Stern S. Comparison of mortality and myocardial infarction rates in angina pectoris with and without ischemic episodes during daily activities. Am J Cardiol 1989;63:273–6.
20. Mulcahy D, Knight C, Patel D, et al. Detection of ambulatory ischaemia is not of practical clinical value in the routine management of patients with angina. A long-term follow-up study. Eur Heart J 1995;16:317–24.
21. Quyyumi AA, Panza JA, Diodati JG, Callahan TS, Bonow RO, Epstein SE. Prognostic implications of myocardial ischemia during daily life in low risk patients with coronary artery disease. J Am Coll Cardiol 1993;21:700–8.
22. Pepin CJ. Ambulatory ischaemia. Eur Heart J 1995;16:297–8.
23. Mulcahy D, Fox K. Ambulatory ischaemia [letter]. Eur Heart J 1995;16:577.
24. Pepine CJ, Cohn PF, Deedwania PC, et al. Effects of treatment on outcome in mildly symptomatic patients with ischemia during daily life: the Atenolol Silent Ischemia Study (ASIST). Circulation 1994;90:762–8.
25. Rogers WJ, Bourassa MG, Andrews TC, et al. Asymptomatic Cardiac Ischemia Pilot (ACIP) study: outcome at 1 year for patients with asymptomatic cardiac ischemia randomized to medical therapy or revascularization. J Am Coll Cardiol 1995;26:594–605.
26. von Arnim T. Medical treatment to reduce total ischemic burden: Total Ischemic Burden Bisoprolol Study (TIBBS), a multicenter trial comparing bisoprolol and nifedipine. J Am Coll Cardiol 1995;25:231–8.
27. Gill JB, Cairns JA, Roberts RS, et al. Prognostic importance of myocardial ischemia detected by ambulatory monitoring early after acute myocardial infarction. N Engl J Med 19[illegible]6;334:65–70.
28. Pepine CJ. Prognostic implica[illegible] of silent myocardial ischemia. N Engl J Med 1996;334:113–4.
29. El-Tamimi H, Davies GJ. Optimal control of myocardial ischaemia: the benefit of a fixed combination of atenolol and nifedipine in patients with chronic stable angina. Br Heart J 1992;68:291–5.
30. Donaldson KM, Dawkins KD, Waller DG. A comparison of nisoldipine and nifedipine, in combination with atenolol, in the management of myocardial ischaemia. Eur Heart J 1993;14:534–9.
31. Treasure CB, Klein JL, Weintraub WS, et al. Beneficial effects of cholesterol-lowering therapy on the coronary endothelium in patients with coronary artery disease. N Engl J Med 1995;8:481–7.
32. Pepine CJ, Geller NL, Knatterud GL, et al. The Asymptomatic Cardiac Ischemia Pilot (ACIP) study: design of a randomized clinical trial, baseline data and implications for a long-term outcome trial. J Am Coll Cardiol 1994;1:1–10.
33. Knatterud GL, Bourassa MG, et al. Effects of treatment strategies to suppress ischemia in patients with coronary artery disease: 12-week results of the Asymptomatic Cardiac Ischemia Pilot (ACIP) study. J Am Coll Cardiol 1994;1:11–20